PanchayatCast: One-Page PRD

Panchayat-level weather intelligence and agro-advisories, downscaled from IMD block forecasts

1. Problem Statement

IMD's agro-meteorological forecasts are issued at block level, yet one block spans dozens of gram panchayats with different terrain, land use and rainfall, so a single block value often leads to wrongly timed sowing, spraying and irrigation. Farmers and extension officers need reliable forecasts and advisories at panchayat scale.

2. Target Users

Persona	Context	Core need
Ramesh - smallholder cotton/paddy farmer, Siddipet (Telangana)	Basic Android phone, reads Telugu, rain-fed plot of 2 acres	"Should I spray, irrigate or sow in the next 2-3 days - in *my* village?"
Lakshmi - Agriculture Extension Officer (AEO) / DAMU staff	Covers 15-25 panchayats, issues advisories twice weekly	Fast, trustworthy panchayat-wise forecasts and ready-to-send advisories

3. Solution Overview

PanchayatCast takes IMD's block-level 5-day forecast and downscales it to every gram panchayat using a **hybrid statistical + ML engine** (physics-based lapse-rate correction, gradient-boosted residual models on terrain/land-use/climatology, and quantile-mapped probabilistic rainfall), kept consistent with the official block forecast. A rule-based crop advisory engine converts the result into actionable, local-language advisories that officers can review on a map dashboard and push to farmers.

4. Success Metrics (Hackathon)

- **Accuracy:** target >= 20% lower temperature RMSE and higher rainfall CSI than the "copy block value" baseline at held-out AWS stations.
- **Coverage:** 5-day forecast for 100% of panchayats in 1 pilot district (approx. 400-500 GPs), 4 variables.
- **Speed:** full daily pipeline (ingest -> downscale -> advisory) runs in under 5 minutes; dashboard loads in under 2 s.
- **Demo:** all 5 Must features shown live in under 90 seconds without errors.

5. Out of Scope

Running our own NWP/WRF model; nationwide rollout; real bulk SMS gateway at scale; crop yield prediction; pest image diagnosis; integration with live IMD internal servers (sample/public feeds are used instead).

6. Assumptions & Risks

Assumption / Risk	Mitigation
Live IMD block forecasts may not be accessible during the hackathon	Use archived IMD samples + open NCEP GFS feed with the same schema; swappable adapter
Sparse ground stations for validation	Train on satellite/reanalysis (ERA5-Land, CHIRPS); reserve AWS stations only for validation
Panchayat-scale rainfall is inherently uncertain	Publish probabilities + High/Medium/Low confidence badge instead of false precision
No long archive of past forecasts for training	Perfect-prognosis training (coarse obs -> fine obs), then apply to live forecasts

7. Tech Stack & Rationale

Layer	Choice	Why
Frontend	Next.js (React) + Tailwind CSS + Leaflet + Recharts	Fast to build, mobile-responsive, strong map + chart ecosystem
Backend / API	FastAPI (Python 3.11)	Same language as the ML stack; async, auto-generated API docs
ML / Geo	xarray, rasterio, geopandas, scikit-learn, XGBoost	Proven, lightweight, explainable; trains on a laptop in minutes
Database	PostgreSQL + PostGIS; Cloud-Optimized GeoTIFFs	Native spatial queries for panchayat polygons and zonal stats
Jobs / Deploy	APScheduler, Docker Compose; Vercel + Render	Simple daily scheduling; one-command reproducible demo